

Supplementary Information

Designing Any Imaging System from Natural Language:
Agent-Constrained Composition over a Finite Primitive Basis

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1 S1. Key Results from the Companion Paper

The companion paper [1] establishes two foundational results that this paper builds upon:

Finite Primitive Basis (FPB) Theorem. Any imaging forward model across 170 modalities and 5 carrier families (photon, X-ray, electron, acoustic, spin/particle) can be decomposed as a directed acyclic graph (DAG) over 11 canonical primitives: P (propagation), M (modulation), Π (projection), F (Fourier encoding), C (convolution), Σ (accumulation), D (detection), S (sampling), W (dispersion), R (scattering), and Λ (nonlinear response). The representation error $\varepsilon_{\text{FPB}} < 0.01$ for all modalities.

Triad Decomposition Theorem. Every reconstruction failure in computational imaging is caused by exactly one of three mechanisms:

- **Gate 1 (Recoverability):** Insufficient measurement information (under-sampling, wrong geometry).
- **Gate 2 (Carrier Budget):** Insufficient signal-to-noise ratio (weak source, noisy detector).
- **Gate 3 (Operator Mismatch):** Forward model does not match the true physics (calibration drift, unmodeled effects).

The total MSE decomposes as $\text{MSE} \leq \text{MSE}^{(G_1)} + \text{MSE}^{(G_2)} + \text{MSE}^{(G_3)}$, guaranteeing diagnostic completeness.

2 S2. Novel System Design Specifications

The 10 novel system designs compose FPB primitives into chains not found in prior literature. Each is specified via the 8-field `spec.md` format. The notation Φ_z denotes depth-parameterised convolution: the C primitive applied with a depth-dependent point spread function $\Phi(z)$, generated by wave propagation through a random phase diffuser with defocus modulation.

2.1 S2.1. 2D Systems

Lensless imaging ($C \rightarrow D$, compression 1:1). A thin random phase mask creates a caustic PSF with broad spatial-frequency support. Reconstruction inverts the convolution via ADMM+TV. PSNR: 43.7 dB, SSIM: 0.984.

```
modality: lensless_imaging
carrier: photon
geometry: single_shot, 128x128
object: 128x128 2D image
forward_model: Convolve(C, psf=phase_mask) -> Detect(D)
```

```

37 noise: Poisson I_0=5000 + Gaussian sigma=3
38 target: PSNR >= 40 dB
39 system_elements:
40   source: broadband LED
41   optics: random phase mask (feature_scale=2.5)
42   detector: CMOS 128x128

```

43 **SIM** ($M \rightarrow C \rightarrow D$, super-resolution). Structured illumination modulates the sample with
44 sinusoidal patterns, shifting high-frequency content into the passband. Three pattern orientations
45 at three phases yield 9 measurements for $2\times$ resolution enhancement. PSNR: 27.7 dB.

46 2.2 S2.2. 3D Systems (8:1 compression)

47 **3D Lensless** ($\Phi_z \rightarrow \Sigma \rightarrow D$). A random phase diffuser creates depth-dependent PSFs through
48 defocus-modulated wave propagation: $\text{PSF}(z) = |\mathcal{F}\{\exp(i\phi_{\text{diffuser}} + i \cdot \text{defocus}(z) \cdot r^2)\}|^2$. Each
49 depth plane produces a unique speckle pattern; the 2D sensor measurement is the sum of all
50 convolved depth planes. This is the C primitive instantiated per depth plane. PSNR: 20.3 dB at
51 8:1 compression.

```

52 modality: 3d_lensless
53 carrier: photon
54 geometry: single_shot, 128x128, n_depths=8
55 object: 128x128x8 3D volume (sparse depth planes)
56 forward_model: Convolve_z(C, psf=diffuser(z)) -> Sum(Sigma) -> Detect(D)
57 noise: Poisson I_0=5000 + Gaussian sigma=3
58 target: PSNR >= 15 dB
59 system_elements:
60   source: broadband LED
61   optics: random phase diffuser (feature_scale=1.0, defocus_max=40)
62   detector: CMOS 128x128

```

63 **Temporal-coded lensless** ($M \rightarrow C \rightarrow \Sigma \rightarrow D$). A binary DMD mask modulates each of 8
64 video frames before diffuser convolution and temporal integration during a single exposure. The
65 mask provides per-pixel temporal diversity. PSNR: 31.6 dB.

66 **Spectral lensless** ($M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$). A coded mask followed by prism dispersion
67 separates 8 spectral bands before diffuser convolution and spectral integration. PSNR: 36.3 dB.

68 2.3 S2.3. 4D Systems (16:1 compression)

69 **4D Spectral-Depth** ($M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Combines coded mask, spectral dispersion,
70 and diffuser depth encoding to recover a (z, λ) datacube ($4 \times 4 = 16$ planes) from a single 2D
71 shot. PSNR: 23.9 dB.

72 **4D Temporal DMD** ($M \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). DMD binary masks provide temporal coding;
73 diffuser PSFs encode depth. Recovers (z, t) datacube at 16:1. PSNR: 25.4 dB.

74 **4D Temporal Streak** ($M \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Passive temporal dispersion (streak camera)
75 replaces active DMD coding. A fixed mask provides spatial diversity. PSNR: 25.3 dB.

2.4 S2.4. 5D Systems (64:1 compression)

5D Full DMD ($M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Full 5D recovery of (x, y, z, λ, t) from a single shot at $4 \times 4 \times 4 = 64:1$ compression using coded mask + spectral dispersion + diffuser depth encoding. PSNR: 29.9 dB.

5D Full Streak ($M \rightarrow W_\lambda \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$). Replaces DMD with passive streak temporal dispersion. The longest chain in the FPB framework (5 operators before detection). PSNR: 29.9 dB.

2.5 S2.5. Design Patterns

Three patterns emerge from the novel designs:

1. **Active modulation (DMD) > passive dispersion:** Binary masks provide per-pixel measurement diversity that continuous dispersion cannot match. DMD-based temporal coding (31.6 dB) outperforms streak-based (25.3 dB) at equal compression.
2. **Algorithm choice is critical for compressive systems:** FISTA+TV or R-L dominate across novel modalities. Inter-method PSNR CoV reaches 42–48% for compressive systems vs. <6% for well-conditioned ones.
3. **Graceful compression degradation:** 1:1 $\rightarrow$ 43.7 dB; 8:1 $\rightarrow$ 20–36 dB; 16:1 $\rightarrow$ 24–25 dB; 64:1 $\rightarrow$ 29–30 dB.

3 S3. Maskless Control Group

To validate that spatial modulation (M) is a prerequisite for compressive dispersion-based imaging, we tested 9 configurations that combine depth encoding (Φ_z) with spectral (W_λ) and/or temporal (W_t) dispersion but *without* a coded mask. All 9 configurations compile through the structural compiler (Gates 1–5) but are flagged by the Judge Agent (Gate 1, recoverability) as severely ill-conditioned:

- Condition number $> 10^6$ for all maskless configurations
- Reconstruction PSNR: 6–9 dB (near noise floor)
- Adding a binary coded mask (M) improves PSNR by 5–15 dB at equal compression

This confirms a fundamental principle: dispersion operators (W) provide only global spectral/temporal diversity, while coded masks (M) provide per-pixel spatial diversity. Both are necessary for well-conditioned compressive recovery.

4 S4. Encoder Comparison for Depth Encoding

Three physically realisable depth-encoding mechanisms were compared across 5 modulation combinations ($3 \times 5 = 15$ configurations total):

Diffuser (Φ_z^{diff}): Random phase plate with defocus-dependent PSFs. $\text{PSF}(z) = |\mathcal{F}\{\exp(i\phi_{\text{random}} + i \cdot \text{defocus}(z) \cdot r^2)\}|^2$.

Multi-lens (L_z): Microlens array producing depth-dependent parallax sub-pixel shifts.

Meta-surface (Ψ_z): Engineered phase pattern (cubic + trefoil + astigmatism aberrations) with defocus modulation.

Key findings:

Table 1: Depth encoder comparison: best PSNR (dB) across 3 encoders $\times$ 5 modulation combinations. $N_z=8$ depth planes, $N_\lambda=4$ spectral bands, $N_t=4$ temporal frames, image size 128×128 .

Encoder	3D only (8:1)	+Mask (8:1)	+M+W $_\lambda$ (32:1)	+M+W $_t$ (32:1)	+M+W $_\lambda$ +W $_t$ (128:1)
Diffuser	17.0	20.3	24.6	26.9	38.1
Multi-lens	16.6	44.1	41.0	36.1	38.7
Meta-surface	16.6	16.2	24.6	25.8	38.0

- Without coded mask: all three encoders perform similarly (16.6–17.0 dB), limited by the ill-conditioning of depth-only recovery at 8:1 compression.
- Multi-lens + mask dominates: parallax shifts are simple, well-conditioned operators (sub-pixel translations). Combined with a binary mask’s per-pixel diversity, the composite forward operator is nearly diagonal, yielding 44.1 dB.
- At maximal compression (128:1, 5D), all encoders converge to ~ 38 dB because the mask and dispersion operators dominate the measurement diversity.

PSF diversity analysis (mean inter-depth cross-correlation):

- Diffuser: 0.29 (moderate diversity, structured speckle)
- Multi-lens: 0.17 (lowest correlation, shift-based diversity)
- Meta-surface: 0.46 (highest correlation, engineered but similar patterns)

5 S5. Reconstruction Details

5.1 S5.1. Algorithms

Five reconstruction algorithms were evaluated for each novel modality:

Wiener deconvolution: Frequency-domain least-squares with noise regularisation. Fastest (<0.1 s) but lowest quality for compressive systems.

GAP-TV: Generalised alternating projection with total variation regularisation [2]. Alternates between data projection and TV denoising.

FISTA+TV: Fast iterative shrinkage-thresholding with TV proximal operator [3]. Nesterov-accelerated gradient descent with $O(1/k^2)$ convergence.

ADMM+TV: Alternating direction method of multipliers with TV splitting [4]. Variable splitting separates data fidelity from regularisation.

Richardson-Lucy (R-L): Maximum-likelihood deconvolution for Poisson noise. Multiplicative updates preserve non-negativity.

5.2 S5.2. Forward Model Implementation

All forward models are implemented as linear operators with explicit adjoint computation. The measurement for a multi-dimensional modality with depth encoding, mask, and dispersion is:

$$y = \sum_{z,\lambda,t} D \left[\text{shift}_{(\Delta x_\lambda, \Delta y_t)} (M \odot \text{conv}(x_{z,\lambda,t}, \text{PSF}(z))) \right] + n$$

where D is detection, M is the binary mask, conv denotes convolution with depth-dependent PSF, shift implements spectral/temporal dispersion as lateral shifts, and n is Poisson + Gaussian

143 noise.

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